Classification of public trust in politicians and regression of cancer mortality rates

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Machine Learning 2: predictive models, deep learning, neural network

Data Science and Business Analytics

Classification project: “Modeling public trust in politicians”

Data description:

European Social Survey (https://ess.sikt.no/en/):

- survey data measuring public attitudes, beliefs and behaviour.
- 11 rounds, 17 datasets

- **CSV data**: 500-600 variables per round (present in one, some or all rounds, may require harmonization across rounds)

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Shapes of datasets:
ESS1 (42359, 567)    ESS7 (40185, 602)
ESS2 (49066, 604)    ESS8 (44387, 535)
ESS3 (47099, 519)    ESS9 (49519, 572)
ESS4 (61009, 674)    ESS10 (37611, 618)
ESS5 (54717, 687)    ESS11 (46162, 676)
ESS6 (54673, 625)
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- **SPSS data** (variable label and value label metadata)

polintr - How interested in politics

How interested would you say you are in politics - are you...

Value	Category
1	Very interested
2	Quite interested
3	Hardly interested
4	Not at all interested
7	Refusal*
8	Don't know*
9	No answer*

*) Missing Value

Data preparation and dataset creation:

1. Literature review and feature selection:

Predictor groups:

- political performance and attitudes (satisfaction with government, democracy, economy, political interest),
- socio-economic conditions (education, income perception, employment status, domicile),
- demographic characteristics (age, gender), contextual identifiers (ESS round and country)
- migration status (born in country, citizenship, parents’ origin),
- religiosity, human values (self-direction, conformity, tradition, equality) and behaviours (trusting others)

2. Variable harmonization across all ESS rounds

round	ESS1	ESS2	ESS3	ESS4	ESS5	ESS6	ESS7	ESS8	ESS9	ESS10	ESS11
essround	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
cntry	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
stfeco	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
stfgov	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
stfdem	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
imwbcnt	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
stfhlth	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
stfedu	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
lrscale	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
polintr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
gincdif	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
eisced	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
hincfel	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
domicil	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
uempla	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

uempli	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
gndr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
agea	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
ppltrst	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
rlgdgr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
rlgdnm	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
brncntr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
ctzcntr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
facntr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
mocntr	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
impfree	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N
ipstrgv	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N
ipfrule	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N
imptrad	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N
ipeqopt	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N
trstplt	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

3. Combining all rounds to one final dataset

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Final shape: (526787, 31)
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Data preparation on final dataset:

Recoding variable-specific missing value codes to NaN using SPSS metadata

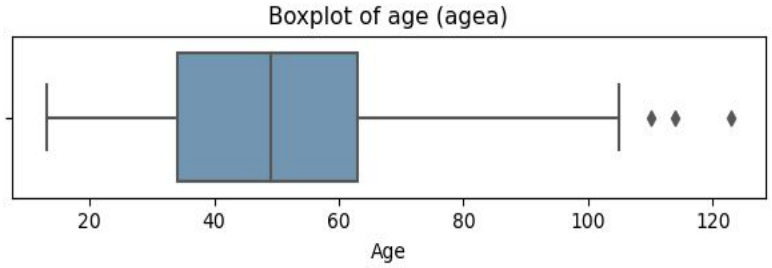
Value	Category
0	Left
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	Right
77	Refusal*
88	Don't know*
99	No answer*

Value	Category
1	Very interested
2	Quite interested
3	Hardly interested
4	Not at all interested
7	Refusal*
8	Don't know*
9	No answer*

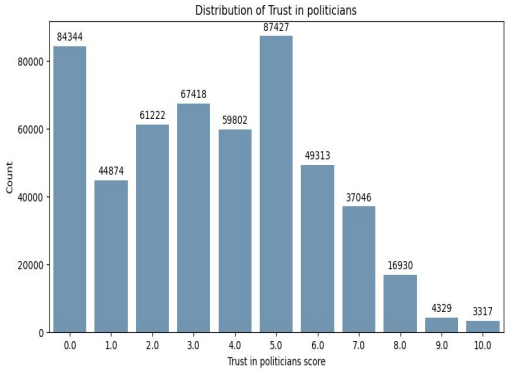
Reducing category sparsity for the country variable by grouping countries into regions

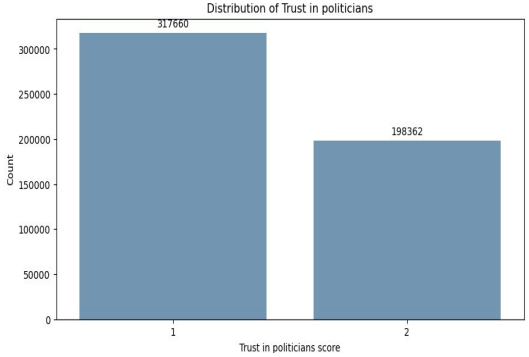
Exploratory data analysis:

Outlier analysis for age:



Target variable distribution and recoding:





Machine learning models

Model setup and preprocessing: Train and test split,
Pipeline for imputing missing values and encoding nominal predictors

Hyperparameter tuning strategy: Tuning is performed on a stratified subsample of the training data to reduce computational cost while preserving class distribution.

Decision tree classifier

- Tree depth limited to 8 to control complexity and reduce overfitting.
- Minimum split and leaf sizes (min_samples_split = 0.003, min_samples_leaf = 0.001) limit overly specific splits, improving model stability and generalization.
- Class weighting (class_weight="balanced") addresses imbalance between low trust and high trust classes.
- Mild cost-complexity pruning (ccp_alpha = 5e-05) slightly improves generalization.

Random forest classifier

- Ensemble of 500 decision trees trained on bootstrapped samples.
- Maximum depth constraint of 15 controls overfitting and improves generalization.
- Feature and sample subsampling (max_features = 0.5, max_samples = 0.5) increases tree diversity and reduces variance.
- Balanced_subsample class weighting corrects class imbalance within each bootstrap sample.

Gradient boosting classifier

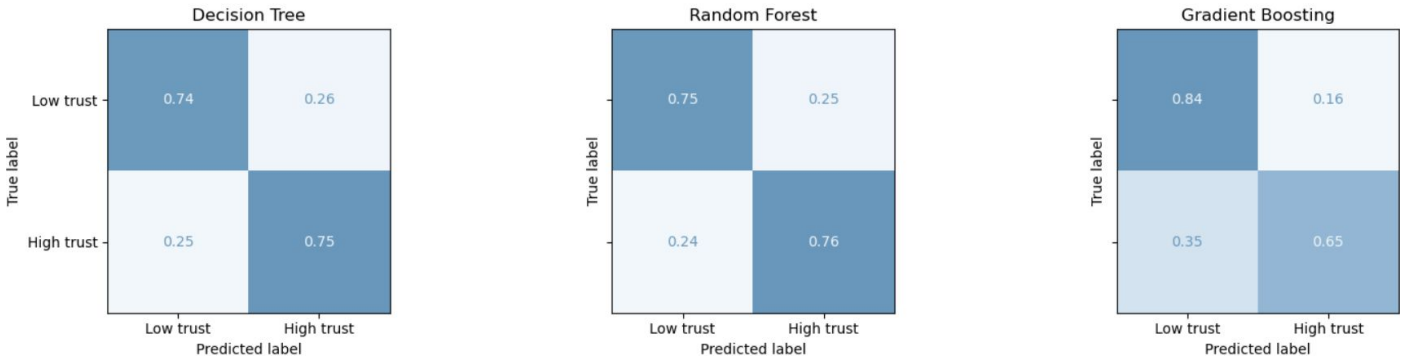
- 400 boosting iterations with shallow trees (max_depth = 3) to limit overfitting.
- Moderate learning rate (0.1) balances gradual error correction and convergence speed.
- The best-performing configuration uses all training observations (subsample = 1.0) and a random subsample of features (max_features = ‘log2’) at each boosting stage.

Model evaluation and comparison

Model performance comparison on the test set

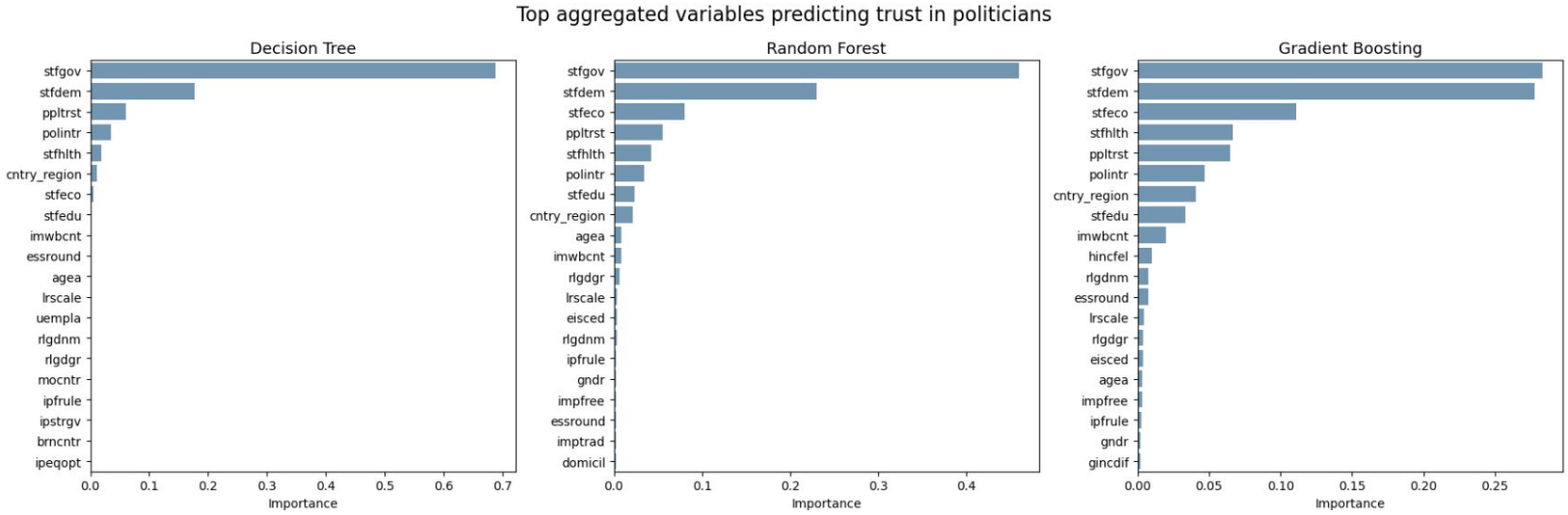
	model	balanced_accuracy	accuracy	precision_class1	precision_class2	recall_class1	recall_class2	f1_class1	f1_class2
1	Random Forest	0.755	0.754	0.834	0.655	0.750	0.761	0.790	0.704
2	Gradient Boosting	0.748	0.770	0.795	0.723	0.844	0.651	0.819	0.685
0	Decision Tree	0.746	0.744	0.827	0.642	0.738	0.753	0.780	0.693

Normalized confusion matrices on the test set



- Decision tree is outperformed by ensemble methods.
- Random forest performs best overall, achieving the highest balanced accuracy (0.755) and the most even recall across classes.
- Gradient boosting has the highest overall accuracy but favors the low trust class - it optimizes total error rather than class symmetry.

Model evaluation and comparison of empirical results with the literature



Summary, conclusions and ethical considerations

- Trust in politicians is driven mainly by institutional performance perceptions, especially **satisfaction with government and democracy**.
- **Regional context and political attitudes matter**, while socio-demographic and human value-related factors play a limited role.
- **Random forest performs best overall**, offering the most balanced and reliable classification under class imbalance.
- Overall, the findings reflect context-dependent patterns in individual perceptions and should be interpreted cautiously for aggregated academic purposes only.